GITAM Deemed to be University

Mechanical Engineering | Course Code: ME304

Heat & Mass Transfer - Comprehensive Syllabus Lecture Notes

Syllabus Focus: ME304 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Heat & Mass Transfer. Includes detailed unit-

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Steady & Transient Conduction

" Syllabus Scope: Fourier law, 3D heat conduction equation in Cartesian and cylindrical coordinates, Crit

Unit 2: Forced & Natural Convection

" Syllabus Scope: Boundary layer approximations, Nusselt, Reynolds, and Prandtl numbers, Empirical correl

ME304: Heat & Mass Transfer - Advanced Topics & Assessment

Academic Track: Mechanical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Thermal Radiation

" Stefan-Boltzmann law, Planck law, Wien displacement law, Emissivity, View factor algebra, Radiation exchange

Unit 4: Heat Exchangers & Mass Transfer

" Overall heat transfer coefficient, Fouling factor, LMTD method, Effectiveness-NTU method, Fick first law of

Unit 5: Boiling, Condensation & Mass Diffusion

" Pool boiling regimes, Critical heat flux, Dropwise and filmwise condensation, Mass transfer coefficients, Sh

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the temperature distribution and heat dissipation from a pin fin of uniform cross-section with
2. Calculate the effectiveness and heat transfer rate of a counter-flow heat exchanger using the Effectiv
3. State reciprocity, summation, and superposition rules for radiation shape factors and find the shape f

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Frank P. Incropera, David P. DeWitt - Fundamentals of Heat and Mass Transfer Wiley 7th Ed
* J.P. Holman - Heat Transfer McGraw-Hill 10th Ed

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